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Empathetic Conversational Agents: Utilizing Neural and Physiological Signals for Enhanced Empathetic Interactions

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ABSTRACT

Conversational agents (CAs) are transforming human-computer interaction, evolving from text-based chatbots to digital humans (DHs) capable of rich emotional expression. This study explores integrating neural and physiological signals into the perception module of CAs to enable real-time emotion detection and empathetic responses. We conducted a user study in which participants engaged with a DH about emotional topics. The DH mirrored participants' emotions in real-time using neural and physiological cues. Results showed that users experienced stronger emotions and greater engagement during interactions with the Empathetic DH, highlighting the benefits of these signals for enhancing empathy. However, challenges remain, including recognition accuracy, emotional transition timing, individual differences, and limited voice modulation. Addressing these issues is key to advancing empathetic digital agents. This research demonstrates the promise of real-time physiological and neural emotion recognition for building emotionally intelligent CAs that foster deeper, more meaningful human-agent interactions.

KEYWORDS

Conversational agent;
neurophysiological emotion
recognition; empathy

1. Introduction

Conversational agents (CAs) are drastically altering the ways we engage with technology and manage our everyday lives in our rapidly digitizing world. A conversational agent in human-machine interaction refers to a desktop agent, virtual character, or robot that is equipped with the capability to engage in human-like conversations with users. For CAs, it is crucial that the machine can give appropriate responses at the content level (relevant and grammatical) as well as the emotion level (consistent emotional expression) (Sun et al., 2018). CAs have shown promising results in various domains, such as intelligent tutoring systems, healthcare, customer services, and virtual therapy (Adikari et al., 2022; D'Mello et al., 2007; Ring et al., 2016; Yalçın, 2018). However, the true potential of CAs lies in their ability to go beyond simple functionality and connect with users on an emotional level. Research suggests that CAs equipped with empathetic communication skills, emotional understanding, and adaptive behavior significantly enhance their effectiveness, intelligence, and reliability (Gratch et al., 2007; Knob et al., 2024; Lucas et al., 2014). This is especially important as CAs begin to move from text-based chatbots to human-like virtual characters capable of rich facial expressions and voice feedback.

Empathy, the ability to understand and share another person's feelings, is a crucial aspect of social relationships (Preston & De Waal 2002). It involves cognitive empathy (understanding others' perspectives), affective empathy (sharing emotions), and somatic empathy (physiological response to others' emotions) (Preston & De Waal 2002). In the context of CAs, creating an empathetic interaction

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involves perceiving and understanding the user's emotional state (cognitive empathy) and delivering appropriate emotional responses (affective empathy) (Allouch et al., 2021; Spitale & Garzotto, 2020). Emotion recognition techniques are vital for CAs to achieve empathetic interactions and become empathetic digital humans (DHs). This includes recognizing emotions from various inputs, such as speech, audio tone, facial expressions, and body gestures (Nandwani & Verma, 2021; Yalçın, 2020). Acoustic signals, for instance, are informative for speech emotion recognition (SER), where algorithms label voice inputs with specific emotional categories (Hu et al., 2023; Zhou et al., 2018). Facial expression recognition is another approach, allowing CAs to adapt verbal and non-verbal behavior based on user emotions (Catania et al., 2023; Khosla et al., 2015). Some research relied on multimodal data and considered audio-visual data to improve interaction. For example, Yalçın, (2020) present a framework for creating empathetic CAs, which can understand and react to others' emotions detected based on audio and video data in real-time.

While audio-visual cues such as speech tone and facial expressions have been widely used for emotion recognition in empathic CAs, they come with important limitations. Facial expressions can be culturally masked or consciously suppressed (Hossain & Gedeon, 2019; Mattioli & Cabitza, 2024), leading to inaccurate emotional inferences across different user groups. Speech and facial recognition also suffer in conditions with poor lighting, occlusion, environmental noise, or privacy constraints (Mattys et al., 2012; Zhao & Li, 2019). Recently, neural and physiological signals have demonstrated reliability in emotion recognition (Ahmed et al., 2023; Blaiech et al., 2013; Shu et al., 2018). Therefore, integrating physiology into the perception module of CAs can meaningfully complement or even surpass the capabilities of conventional visual and auditory channels, particularly in contexts requiring deeper affective insight or robust sensing under challenging conditions. However, existing research has yet to exploit these cues in the perception module of Empathetic DHs. In this paper, we propose to use neural and physiological cues in the perception module of CAs to achieve more accurate empathy responses. These modalities can be especially valuable when behavioral cues, like facial expressions or speech, are difficult to measure, like in noisy and dark environments.

The primary focus of this paper is to investigate the effectiveness of integrating neural and physiological signals into the perception module of a conversational agent (CA). Additionally, it evaluates the efficiency of physiology-based multimodal emotion recognition in enhancing the empathy of virtual conversational agents. We describe the CA's components and propose the design for the perception module to foster empathy between the CA and users. A user study is conducted to evaluate the proposed CA, and the results are presented and discussed, along with potential avenues for future research.

The main contributions of this paper are as follows:

- Using neural and physiological signals in the perception module of a conversational agent to make an empathetic interaction.
- Creating a publicly available dataset of multimodal data, including facial video, recorded audio, and neural and physiological signals in interaction with empathetic and neutral conversational agents¹.
- Identifying the challenges and limitations of real-time interaction with conversational agents that use neural and physiological signals.

2. Related work

2.1. Advancements in affective computing and emotion recognition

Due to their accessibility and ease of use, current methods of emotion recognition predominantly employ behavioral signals such as facial expressions (Li & Deng, 2022; Samadiani et al., 2019; Sun et al., 2020), text, and audio analysis (Sailunaz et al., 2018). These modalities are commonly used in many human-computer interaction (HCI) applications (Samadiani et al., 2019). Some applications, like gaming (AlZoubi et al., 2023), self-driving cars (Siam et al., 2023), online embedded systems (Jeong & Ko, 2018), and social robots and conversational agents (Antony et al., 2021; Luo et al., 2022; Ruiz-Garcia et al., 2018), use real-time facial expression recognition and speech analysis to

understand the user's state. Although these approaches display potential in identifying basic emotions, they encounter obstacles such as cultural disparities, diversity in age and gender, and the susceptibility of behavioral modalities to manipulation by the user, thereby allowing for easy deception (Hossain & Gedeon, 2019).

Researchers have increasingly incorporated physiological measures like Electroencephalography (EEG) (Alarcao & Fonseca, 2017), Galvanic Skin Response (GSR), heart rate variability (HRV) (Udovičić et al., 2017), blood volume pressure (BVP), body temperature (BT), respiration rate (RR), etc. (Chunawale et al., 2020) to address these limitations for emotion recognition. Many researchers have shown that although EEG and physiological signals are weak, they can be used to reliably recognize underlying emotions (Blaiech et al., 2013; Shu et al., 2018). Combining these various modalities can further improve the accuracy of emotion recognition (Huang et al., 2019; Zhu et al., 2020). Despite advancements in neural and physiological emotion recognition, these modalities have rarely been used in real-time emotion recognition. Their potential and limitations need to be explored more deeply for real-life scenarios such as interaction with conversational agents.

2.2. Empathetic conversational agents

The evolution of conversational agents towards achieving empathetic interactions has seen significant advancements in the past few years. Empathetic conversational agents or affective conversational agents are designed to understand and respond to users in a way that reflects an understanding of the user's emotions, providing a more human-like interaction experience (Smith et al., 2011; Yang et al., 2019). Samrose et al. (2020) explored the potential of empathetic conversational agents in mitigating boredom, highlighting the importance of emotional intelligence in conversational agents to adjust user mood and enhance task performance. Sakhrani et al. (2021) have focused on creating empathetic agents for mental health support, emphasizing the need for agents that can conduct empathetic and free-flowing conversations to assist individuals who may be hesitant to seek help from human therapists.

Most conversational agents rely only on text emotion recognition, which is a widely studied area (Antony et al., 2021; Luo et al., 2022; Provoost et al., 2017). For example, Zaranis et al. (2021) developed EmpBot, an empathetic chatbot that utilizes sentiment understanding and empathy forcing to enhance the empathetic responses of conversational agents. Ayshabi and Idicula (2021) introduced a model that employs a multi-resolution mechanism with multiple decoders to capture the nuances of user emotions and generate empathetic responses. This approach aims to improve the perception and expression of emotional states in dialogue systems. However, facial expressions and speech tone have more information about the user's emotion (Luo et al., 2022). For example, Hussain et al. (2012) presented a novel architecture for affective interaction with conversational agents. They fused audio, video, and text to detect affective states. Tanaka et al. (2017) introduced a social skill training agent for people with Autism that uses audio-visual features for detecting emotional states. Similarly, Aneja et al. (2021) introduced a conversational agent that can recognize human behavior during open-ended conversations and automatically align its responses to the conversational and expressive style of the other party by using multimodal inputs, including facial expressions and speech. Tavabi (2019) developed a multimodal deep neural network that integrates emotional tone in language and facial expressions to recognize opportunities for empathetic responses in human-agent interactions. This study highlights the potential of incorporating visual and auditory cues to enhance the emotional intelligence of Empathetic CAs.

Despite these advancements, a significant gap remains in integrating the neural and physiological data spectrum for emotion recognition. While multimodal approaches have begun to incorporate visual and auditory cues, the potential of neural and physiological signals (e.g., EEG, HRV, and EDA) in enhancing the empathetic capabilities of conversational agents remains largely untapped. These data sources could offer deeper insights into the user's emotional state and reveal genuine emotions reliably (Hossain & Gedeon, 2019; Shu et al., 2018). This enhanced understanding would empower CAs to generate more nuanced and authentically empathetic responses. However, conversational agents rarely use neural and physiological modalities to recognize emotion in real-time. One exception was the work of Val-Calvo et al. (2020), who used an affective robot for storytelling and created a realistic paradigm for

evoking emotions. They recorded EEG, BVP, GSR, and video data and estimated emotional state in real-time subject-dependently. However, the robot did not use the recognition result to adapt its behavior. Similarly, Ghandeharioun et al. (2019) introduced EMMA, which is an emotion-aware chatbot that detects emotions in real-time based on the user's smartphone location and provides wellness suggestions for mental health according to detected emotions using location data. However, they did not consider any other source of data, especially physiological cues. Equipping conversational agents with physiological emotion recognition is crucial, especially in scenarios where recording audio and video is challenging due to environmental factors like noise, poor camera angles, or low light conditions. Moreover, audio and video alone often fail to fully capture emotional states. For example, when supporting older adults, individuals with autism, or those who face difficulties expressing emotions verbally or behaviorally, leveraging neural and physiological cues can provide a more accurate and effective understanding of their emotional needs.

3. Methodology

3.1. Conversational agent design

We used a digital human (DH) developed by Soul Machines Ltd² (Figure 1) to create an empathetic conversational agent. We selected Soul Machines' digital human platform as the basis for our virtual agent due to its advanced capabilities in autonomous facial expressions and real-time responsiveness, features critical to our study on empathy and emotion mirroring. Unlike MetaHumans³, which are primarily optimized for high visual fidelity in gaming and cinematic environments, Soul Machines is purpose-built for emotionally engaging interactions, with applications in healthcare, education, and customer service. This made it particularly suitable for our research goals, which prioritize emotional expressiveness over purely conversational interaction. Additionally, an Orchestration Server is an additional layer of logic that manages real-time interactions between users and Digital Humans by coordinating conversation flow, synchronizing speech and expressions, and integrating with external systems. It can also call external code or services for real-time analysis, such as emotion detection, data retrieval, or decision-making, allowing the Digital Human to respond in a context-aware and emotionally intelligent way.

While the version of the platform used in this study did not include support for large language models (LLMs), verbal interaction was not a key focus of our experiment. Instead, we used pre-defined answers and concentrated on neurophysiological emotional responses and their impact on perceived empathy.



Figure 1. The soul machine's DH that we used in this study.

The DH was represented as a young adult female. The avatar's face was chosen from the available options to align with the predominant local cultural context, ensuring consistency and minimizing variability in participant responses. We disabled all of the Soul Machine features of speech and facial expression recognition so that we could implement our own.

We created two different designs of DH in this research. One is a neutral DH with a simple perceptual module. It did not have an emotion recognition module and responds with a neutral pre-programmed answer, and takes appropriate action based on the user's input and dialog flow. The other is an empathetic DH whose perceptual module is based on neural and physiological signals. It recognizes emotions in real-time and creates the appropriate expression by mirroring detected emotions. It chooses the appropriate response from a dataset of responses that reflect and appreciate the user's detected emotions. We used a female DH whose appearance and voice were identical in both DHs.

3.2. Acquiring human emotional responses

In this study, we employed the Octopus-Sensing software suite (Saffaryazdi et al., 2022) to gather data from multiple sensors and perform real-time monitoring simultaneously across all of them. The data acquisition included facial video, audio, Electroencephalography (EEG), electrodermal activity (EDA), and photoplethysmography (PPG). The EEG was captured by the OpenBCI EEG cap⁴, while the EDA and PPG signals were captured using the Shimmer3 sensor⁵. Participants began by providing informed consent and receiving instructions. They were then seated in front of a monitor and laptop and fitted with an OpenBCI cap and Shimmer sensor in their non-dominant hand. Participants were alone during the data collection period in a quiet, controlled laboratory setting. The experimental setup and the devices used for data collection are shown in Figure 2. The acquired EEG, PPG, and EDA signals were primarily used for real-time emotion recognition. Although audio and video were not used in this paper, they were collected for potential future research.

3.3. Architecture

We developed an orchestration server, which acts as the Soul Machine orchestration layer, to perform several functions within the empathetic DH. The server's main tasks included understanding the user's intent, selecting the most suitable pre-written response from a database of responses provided in the supplementary materials, and generating appropriate facial expressions. It controls the dialog flow and creates responses based on the detected emotions through a real-time emotion recognition system. The orchestration server is also responsible for regulating the expressions of the digital human based on the user's real-time emotional state. Figure 3 shows the overall architecture of the empathetic DH that works based on neurophysiological responses.



Figure 2. The experimental setup and the devices used for data collection.

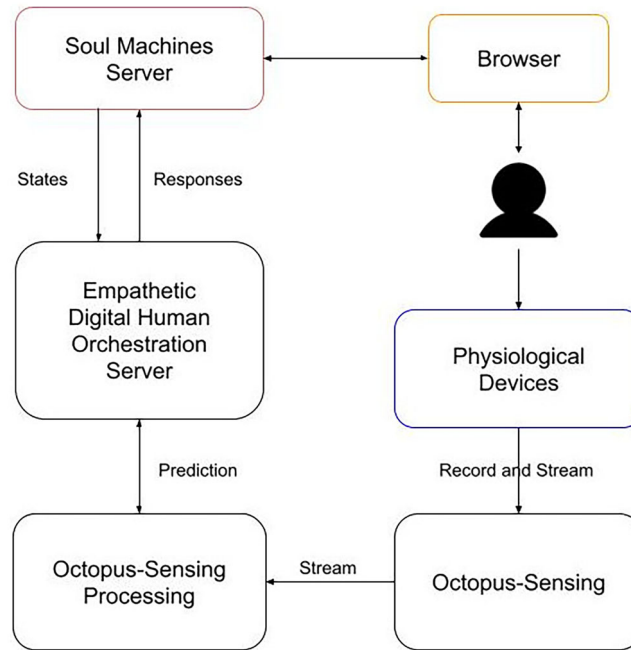


Figure 3. The overall architecture of the embodied conversational agent (ECA).

We customized the appearance of the DH and disabled all emotion recognition features available through the Soul Machines web browser. Instead, we employed our own multimodal emotion recognition system and integrated it with the DH using an orchestration server. In the empathetic condition, the real-time emotion recognition system was activated every 5 s, analyzing emotions using data from the previous 20 s. After each detection, which occurred every 5 s, the DH adjusted its facial expressions to reflect the recognized emotions. Additionally, during interactions, the DH responded according to the most frequently detected emotion during the user’s speech.

In this study, we used the dataset from prior research (Saffaryazdi et al., 2024) to train emotion recognition models. This dataset includes EEG, PPG, and EDA data collected from 16 participants during dyadic conversations conducted in both remote and face-to-face settings, with each condition consisting of 8 trials. Due to technical issues, particularly with EEG signal recording in this dataset, some data were missing. Consequently, we had 240 trials to train the EDA and PPG models and 204 trials to train the EEG model. Since this dataset’s labels are based on a dimensional arousal-valence model and earlier research explored that dimensional emotion models are more sufficient, reliable, and more accurate for self-assessments in explaining emotions (Eerola & Vuoskoski, 2011; Lichtenstein et al., 2008), and dataset labels are based on an arousal-valence model, we used a 2D-dimensional emotion model, and the results of predictions are based on arousal and valence. Following established practices in affective computing and emotion recognition research (Galvão et al., 2021; Koelstra et al., 2012; Zitouni et al., 2023), we transformed participants’ 5-point affective ratings—collected via the Self-Assessment Manikin (SAM) questionnaire (Bradley & Lang, 1994), into binary labels representing arousal (low vs. high) and valence (negative vs. positive). Specifically, the five-point scales for arousal and valence were mapped to binary values of -1 (low/negative) and 1 (high/positive), as defined by Equations (1) and (2). In the affective computing field, arousal and valence are often measured on continuous or multi-point scales to capture nuanced emotional states. However, converting these to binary values is a common and pragmatic approach to reduce model complexity, simplify classification tasks, and improve computational efficiency, particularly when working with limited data, subjective annotations, or targeting real-time, robust emotion detection in applied settings (Koelstra et al., 2012; Marín-Morales et al., 2018; Zitouni et al., 2023). Binary classification also helps mitigate class imbalance and individual differences that may limit generalization.

We followed the same methodology as described in Saffaryazdi et al. (2024) to preprocess data, extract features, and train emotion recognition models using each signal. This approach has proven

effective in recognizing emotions during conversations. We extracted features listed in Table 4 and used the Random Forest Classifier (RFC), previously employed in emotion classification research (Menezes et al., 2017; Chaparro et al., 2018), to train the models for real-time emotion recognition.

$$Arousal = \begin{cases} -1 & \text{if } ArousalRate \leq 3 \\ 1 & \text{otherwise} \end{cases} \quad (1)$$

$$ArousalRate = \{1, 2, 3, 4, 5\}$$

$$Valence = \begin{cases} -1 & \text{if } ValenceRate > 3 \\ 1 & \text{otherwise} \end{cases} \quad (2)$$

$$ValenceRate = \{1, 2, 3, 4, 5\}$$

The emotion recognition system employs a sliding-window approach, in which emotional state predictions are updated every 5 s using the most recent 20 s of multimodal physiological data (EEG, EDA, and PPG). In this setup, a new 20-second data window is analyzed every 5 s—resulting in overlapping windows—to continuously track the user’s emotional state. This design was not driven by system performance constraints, but rather was empirically determined to optimize both signal fidelity and system responsiveness. The 20-second window length was selected based on the physiological requirements of the modalities involved. In particular, PPG-derived features such as heart rate variability (HRV) necessitate a minimum window length of approximately 20 s to allow for reliable feature extraction. Additionally, this window provides a temporally stable context that helps smooth short-term fluctuations in EEG and EDA signals, improving classification robustness. The 5-second step size strikes a balance between temporal granularity and latency. It allows the system to detect and reflect emotion changes in near real-time while maintaining sufficient overlap for continuity in signal trends. This interval was empirically chosen to support fluid, emotionally responsive interactions with the digital human (DH).

We used the weighted decision fusion method shown in Equation (3) to combine the recognition results from neural and physiological modalities and detect the emotional state. Equal weights were considered for detecting valence levels for all modalities’ predictions. However, we assigned a higher weight to EDA and PPG for arousal detection, setting their weight at 2, while EEG was weighted at 1. Our primary objective in this research was to enhance real-time recognition accuracy based on established signal reliability, so we made this decision based on evidence showing that PPG and EDA signals have greater reliability in detecting arousal levels (Kołodziej et al., 2019; Tarnowski et al., 2018; Udovičić et al., 2017). The Equation (4) shows how we mapped the result of fusion to −1 and 1 values. The empathetic DH expressed emotions based on the result of the fusion method, primarily by mirroring the emotional state as listed in Table 1.

$$p_o^x = a \times p_{EEG}^x + b \times p_{GSR}^x + c \times p_{PPG}^x \quad (3)$$

$$x \in [Arousal, Valence]$$

$$a = b = c = 1 \text{ for Valence}$$

$$a = 1, b = c = 2 \text{ for Arousal}$$

$$Fusion = \begin{cases} -1 & \text{if } p_o^x < 0 \\ 1 & \text{otherwise} \end{cases} \quad (4)$$

$$x \in [Arousal, Valence]$$

To map the result of predictions to the avatar expressions, we used Table 1. In this study, we intentionally limited the range of modeled emotions to simplify the problem space and assess the feasibility of real-time emotional mirroring via a virtual agent using neurophysiological signals. Our emotion

Table 1. Defining the empathetic DH expressions according to the result of the fusion method.

Arousal	Valence	State	DH expression
1	1	HAHV	Strong happiness
1	−1	HALV	Anger/Scaredness
−1	1	LAHV	Slight happiness
−1	−1	LALV	Sadness

classification was based on the arousal-valence framework, which organizes emotions according to their physiological activation (arousal) and emotional positivity or negativity (valence). As shown in Table 1, we mapped positive emotions at two arousal levels to happiness (slight and strong), low-arousal negative emotions to sadness, and high-arousal negative emotions to a combined category of anger and fear.

This selection was guided by two main considerations. First, these emotions are among the most distinguishable within the arousal-valence space—happiness is high-valence, sadness is low-arousal, and both anger and fear are high-arousal and low-valence. However, since the conversation topics did not include fear-inducing content, most low-valence, high-arousal signals were interpreted as anger. Grouping anger and fear reduced classification ambiguity, with anger being the dominant interpretation in this context.

Second, we considered the expressive capabilities of the Soul Machines avatar. The platform effectively rendered happiness and sadness, with two expression intensities for happiness that aligned well with our arousal-based model. In contrast, its negative facial expressions were less distinct across emotion types, making differentiation between anger and fear challenging. As a result, the avatar expressed negative states through compassionate verbal cues rather than differentiated facial expressions. While this approach limited emotional granularity, it ensured clarity and robustness for this early-stage system. Future work will expand the emotional range as recognition models and avatar expressiveness improve.

4. User study design

We conducted a within-group user study to investigate the empathy between a human and a DH when the DH behaves emphatically compared to when the DH behaves in a neutral way. The study procedure was approved by and performed in accordance with the guidelines and regulations of the University of Auckland Human Participants Ethics Committee on March 2, 2020 (reference number: 023799). All participants were fully informed about the scope of the study, including potential risks and discomfort associated with wearing physiological sensors (EEG, EDA, and PPG). Detailed information was provided regarding the type of data collected, how it would be stored and used, and participants' rights concerning their data. Informed consent was obtained from all participants prior to the study, emphasizing their right to withdraw at any time without penalty. Data privacy was prioritized through anonymization and secure data handling protocols approved by the university ethics committee.

4.1. Procedure

We recruited 23 participants aged between 21 and 44 years old ($\mu = 30, \sigma = 6$). The participants were university students and staff, and they were offered a \$40 voucher as an incentive for participating in the study.

The experiment began by explaining the goals, devices, and steps and completing a consent form and demographic questionnaire. All participants had a set of conversations with both DHs and talked about different emotional topics, alternating the start condition (conversation with neutral or empathic DH) and the topic (image set) in a counterbalanced way. In the neutral condition, the DH asked questions, responded neutrally, and showed no expression when the participant was talking. In the empathetic condition, the DH reflected the recognized emotion in its expressions and responses. The empathetic DH expressed its feelings about the image, asked questions, and responded in an empathetic way according to the participant's response.

We followed the approach described in Saffaryazdi et al. (2024) to generate emotional conversations but replaced the human interviewer with the DH. We selected eight images from the IAPS image set (Lang & Bradley, 2007) as the topic of conversation, two from each of the following categories: High Arousal-High Valence (HAHV), High Arousal-Low Valence (HALV), Low Arousal-High Valence (LAHV), and Low Arousal-Low Valence (LALV). We sorted the dataset according to arousal and valence level, ignored images with high negative subjects like suicide or violence, and chose images

with the highest and lowest arousal and valence levels with popular subjects to discuss. After displaying an image from the IAPS dataset, the DH started conversing with the participant about their feelings about the image and related experiences.

4.2. Emotion and empathy self-report data

After each conversation topic, all participants were asked to complete a self-report questionnaire which was designed according to the definition of emotion and empathy and previous studies (Jauniaux et al., 2020; Koelstra et al., 2012; Witchel et al., 2016). This data was used to assess real-time emotion recognition and the level of empathy between the participants and the digital human agent. In addition, we used the self-report data to evaluate whether the interaction with the digital human could evoke emotions in the participants.

The questionnaire consisted of 8 questions, categorized into three parts. The first part (questions 1 and 2) aimed to evaluate the participants' emotional state based on the 2-dimensional emotion model. We compared their responses to these questions with the results of real-time emotion recognition for each trial to assess the quality of the emotion recognition.

The second part (questions 3–5) was created according to the definition of empathy and previous studies to evaluate empathy (Jauniaux et al., 2020; Witchel et al., 2016). They were used to evaluate the perceived level of empathy displayed by the digital human agent towards the participants, including cognitive and affective empathy.

The third part (questions 6–8) focused on assessing the appropriateness and timing of the digital human's emotions and expressions. These questions were designed to gauge how well the DH's emotions and expressions aligned with the participant's emotions. Table 2 provides an overview of the questionnaire and its related factors.

4.3. Human-agent rapport

To measure the participants' feelings about their interaction with the agent, we used the post-interaction questionnaire after each condition, adapted from the questionnaire used in earlier research (Cerekovic et al., 2014; Heerink et al., 2010). As shown in Table 3, this questionnaire featured statements that could be responded to on a 5-point Likert scale. This questionnaire evaluates the quality of interaction (QoI), degree of rapport (DoR), degree of liking the agent (DoL), and degree of social presence (DSP). The first three factors were derived from the perception of interaction questionnaire developed by Cerekovic et al. (2014). They have been inspired by the study (Cuperman & Ickes, 2009), in which the authors examine how the Big Five personality traits are expressed in mixed-sex dyadic interactions between strangers. To measure perceived interaction, they developed a "Perception of Interaction" questionnaire containing items that evaluate various aspects of participants' interaction experiences. The last factor, DSP, has been validated by Heerink et al. (2010) with Cronbach's alpha of 0.83, indicating good reliability. Table 3 shows the details of the human-agent rapport questionnaire.

Table 2. Self-report questionnaire after each conversation.

Category	Question	Variable
Emotional state	1- How negative or positive was the emotion that you felt?	Valence
	2- What was your arousal level: calm to excited?	Arousal
Human-agent empathy	3- I felt that DH was empathetic toward me.	Empathic
	4- DH identified what I was feeling.	Cognitive
	5- DH reflected on my emotions.	Affective
Real-time expressions	6- The emotion that DH displayed was appropriate.	Appropriate_emotion
	7- DH displayed the best expression for the situation.	Best_expression
	8- DH's emotional expressions occurred at appropriate times.	Appropriate_time

DH: digital human.

Table 3. Human-agent rapport questionnaire and the category of each question.

Question	Category
The interaction with DH was smooth, natural, and relaxed.	QoI
I felt accepted and respected by DH.	DoR
I think DH is likable.	DoL
I got along with DH pretty well.	DoR
I did not want to get along with DH.	DoL
The interaction with DH was forced, awkward, and strained.	QoI
I felt uncomfortable during the interaction.	QoI
I felt that DH was paying attention to my mood.	DoR
I was paying attention to the way that DH responded to me and I was adapting my own behavior to it.	DoR
I think DH finds me likable.	DoL
When interacting with DH, I felt like interacting with a real person	Sp
I enjoyed the interaction.	QoI
I sometimes felt like DH was actually looking at me	Sp
I would like to interact more with DH in the future.	DoL
I can imagine DH as a living creature	Sp
DH often said things completely out of place.	QoI
The interaction with DH was pleasant and interesting.	QoI
I often realized DH is not a real living creature	Sp
Sometimes it seemed as if DH had real feelings	Sp

The categories include quality of interaction (QoI), degree of rapport (DoR), degree of liking the agent (DoL), and degree of social presence (DSP).

5. Analysis

5.1. Self-report data

Since all participants had conversations with neutral and empathetic DHs, we performed a within-group statistical analysis to see the effect of DH type on humans. Data were analyzed using Python and the ARtool package (Kay & Wobbrock, 2016) in R. The self-report data were based on a 5-point Likert scale, so our data were analyzed using non-parametric methods. We used the aligned rank transform (ART) with repeated measures ANOVA (Wobbrock et al., 2011) to statistically analyze non-parametric data. A set of three-way repeated measures ANOVA with ART was conducted to examine the effect of DH type (neutral or empathetic) and emotional state of conversation on empathy, emotional state, and the appropriateness of DH responses and expressions. We transferred the reported arousal and valence levels to the low(−1) and high(1) values according to Equations (1) and (2) to reduce the complexity and increase the recognition process in real-time. Additionally, using continuous or multi-class labels can lead to data sparsity, complicating the training of robust models. Low and high labels can address this issue by consolidating data into broader categories, making the classification results more interpretable for end-users. We considered low and high arousal and valence values as the emotional state of conversation for statistical analysis.

Similarly, to statistically analyze the effect of agent type on the human-agent report, we conducted a set of one-way repeated measures ANOVA with ART when the agent type was the condition.

5.2. Human body responses

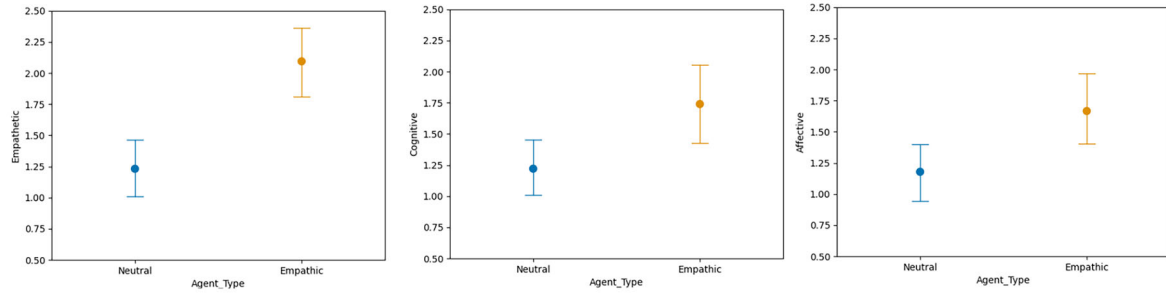
To compare the effect of agent type on human body responses, we extracted some common and descriptive features from the cleaned EEG, PPG, and EDA signals as described in Table 4. Since these data were not normally distributed, we used a one-way repeated measures ANOVA with ART when the agent type was the condition.

6. Results

To evaluate the agent's empathy, we assessed the three components of empathy including cognitive empathy, affective empathy and somatic empathy (Preston & De Waal, 2002). First, statistically show the human-agent empathy and rapport based on self-report data (Cognitive empathy). Second, we examine the differences in human neural and physiological responses when interacting with neutral

Table 4. Features extracted from the EEG, EDA, and PPG signals.

Modality	Features	Feature details
EEG	Power spectral density (PSD) of EEG band powers	psd_theta, psd_alpha, psd_beta psd_gamma, psd_delta
PPG	Some HRV time domain features using neurokit library	HRV_MeanNN, HRV_SDNN, HRV_RMSSD, HRV_SSD, HRV_CVNN, HRV_MedianNN, HRV_CVSD, HRV_MadNN, HRV_MCVNN, HRV_IQRNN, HRV_pNN50, HRV_pNN20, HRV_HTI, HRV_TINN
EDA	Some statistics of phasic and tonic components using neurokit library	mean, standard deviation, and variance of phasic and tonic components and peaks

**Figure 4.** Effect of agent type on empathy factors, including cognitive and affective empathy and overall empathy level.**Table 5.** Effect of emotional state on empathy factors in the empathetic DH.

Variable	Condition	<i>F</i>	<i>Df</i>	<i>Df.res</i>	<i>p</i>
Empathetic	Arousal	17	1	84	<0.001
Empathetic	Valence	23	1	87	<0.001
Cognitive	Arousal	29	1	91	<0.001
Cognitive	Valence	26	1	91	<0.001
Affective	Arousal	29	1	90	<0.001
Affective	Valence	25	1	91	<0.001

versus empathetic avatars (Somatic empathy). Finally, we assess the quality of emotion recognition and real-time expressions (Affective empathy).

6.1. Human-agent empathy

We conducted a series of one-way repeated-measure ANOVA with ART to examine the empathy between humans and both DHs. The assessment of empathy was based on three reported factors: (1) the overall empathy level between the human and agent (empathy), (2) the agent's ability to identify emotions (cognitive empathy), and (3) the agent's capacity to reflect emotions (affective empathy). The results of the analysis revealed significant differences in all of these factors, overall empathy ($F(1, 166) = 27, p < 0.001$), cognitive empathy ($F(1, 166) = 4.7, p < 0.03$), and affective empathy ($F(1, 166) = 5.4, p < 0.02$) during interactions with both neutral and empathetic DHs.

Figure 4 illustrates the empathy factors based on the type of agent. It is apparent from the figure that the average scores of the empathy factors were significantly higher in conversations with the empathetic DH in comparison to the neutral DH. However, the ratings were not notably high on average. To understand the reasons behind these moderate ratings, we further evaluated the effect of participants' emotional state during the conversation on the empathy factors.

Similar to the previous section, we used a set of two-way repeated-measure ANOVA with ART to examine how the emotional state affects empathy factors. The results of the ANOVA test revealed significant differences in empathy factors across various emotional arousal and valence levels (Table 5). However, there were no significant interactions between arousal and valence levels.

As demonstrated in Figure 5, the ratings for all three factors were remarkably higher during positive emotions and high arousal states. This result aligns with the result in the previous section, indicating that in high valence and arousal, because reflected emotions are more appropriate and at an appropriate time, people feel more empathy during their interaction with the DH.

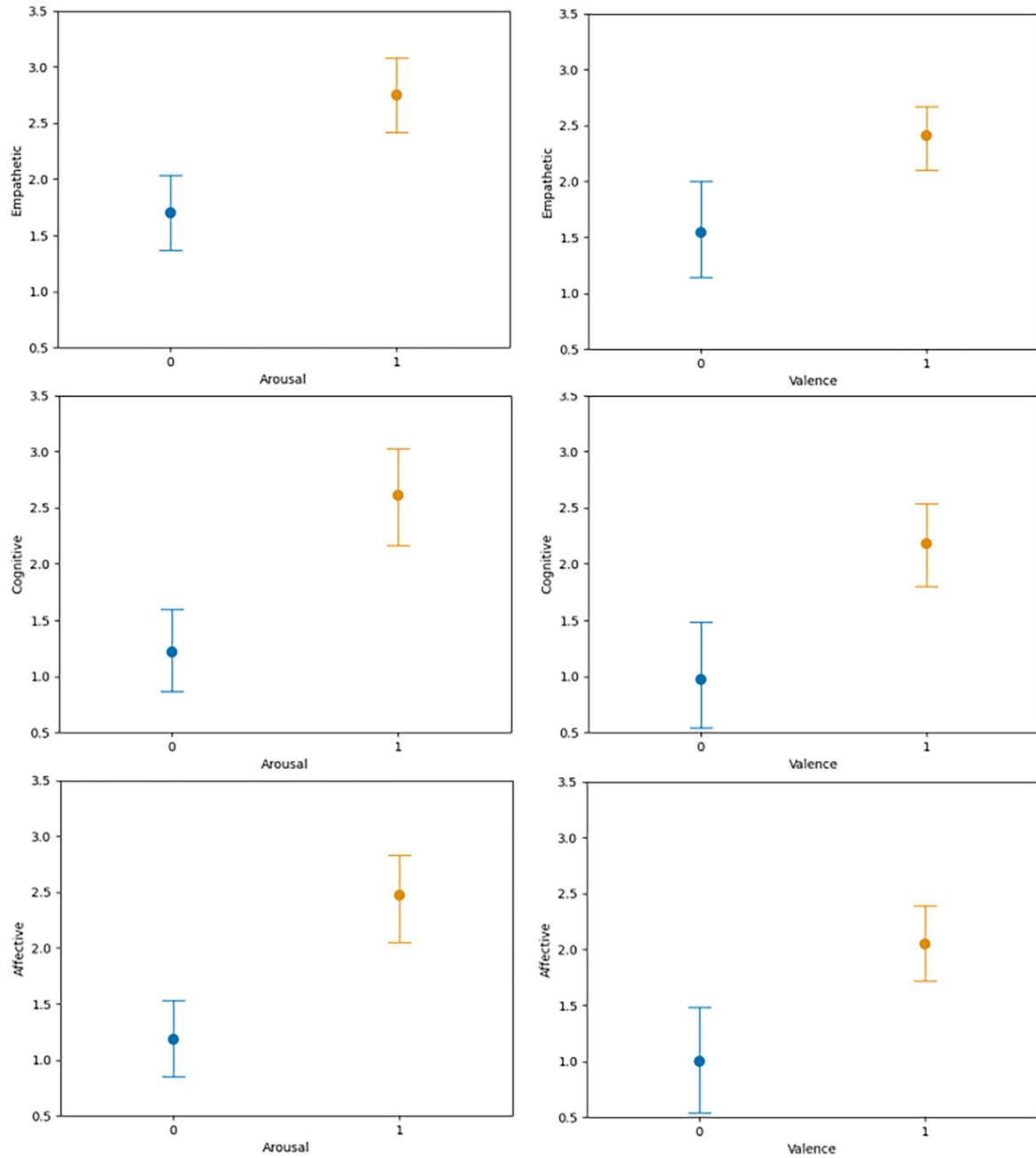


Figure 5. Effect of the emotional state of conversation topic on empathy factors.

6.2. Human-agent rapport

A series of one-way repeated measures ANOVA was conducted to examine the effect of agent type on human-agent rapport factors. Results showed that the type of agent led to statistically significant differences in the Degree of Rapport (DoR) ($F(1, 42) = 8.38, p = 0.006$) and DoL ($F(1, 42) = 6.64, p < 0.01$). The other two factors, including Degree of Social Presence (DSP) and Quality of Interaction (QoI), were not significantly different. However, point plots in Figure 6 show a higher rating in the interaction with the empathetic agents.

We also measured the percentage of participants who reported a higher level of DoR, DoL, DSP, and QoI in interaction with the empathetic agent compared to the neutral one. For all factors, a higher portion of participants reported higher levels of DoL, DoL, DSP, and QoI in interaction with the empathetic agent, which were 82, 86, 59, and 63%, respectively. The higher percentage of participants

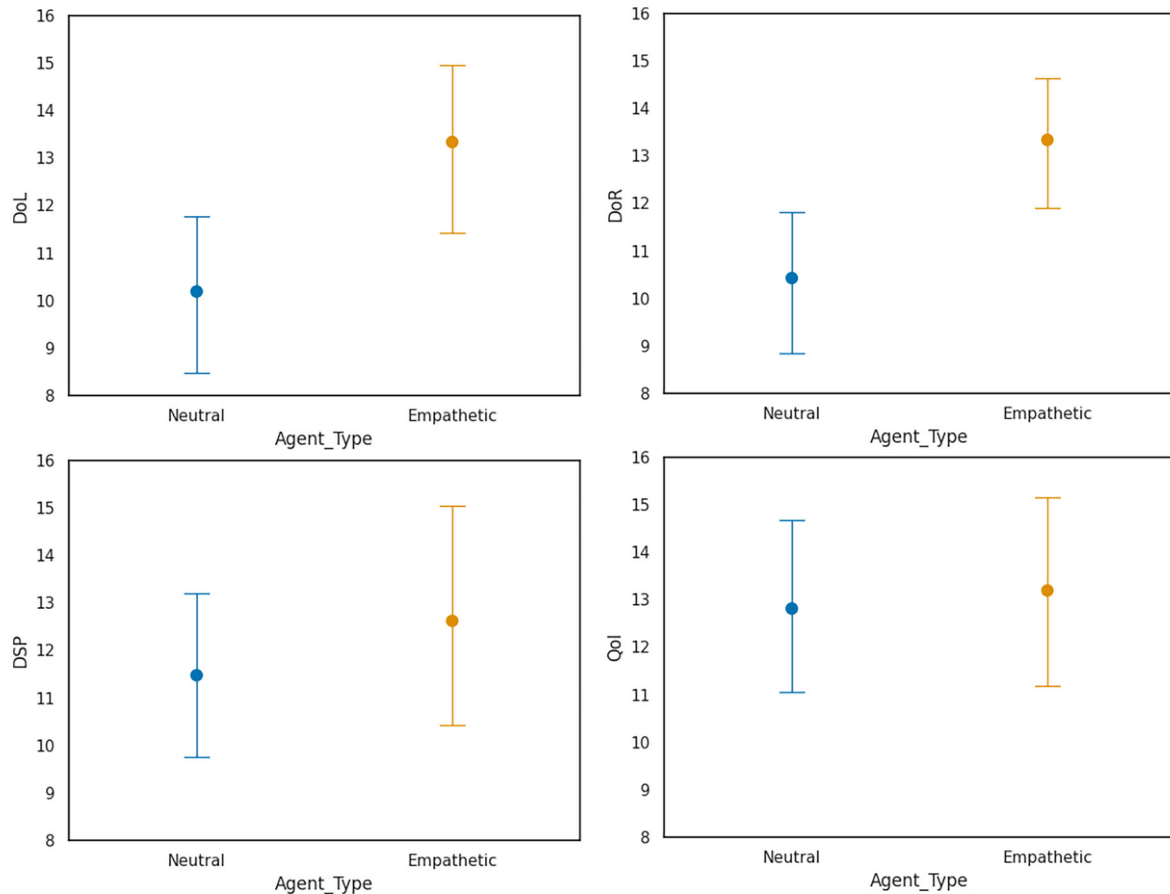


Figure 6. Point-plots of human-agent rapport factors. DoL: Degree of liking; DoR: Degree of rapport; DSP: Degree of social presence; QoI: Quality of interaction.

who reported higher levels of DoR, DoL, DSP, and QoI in interaction with the empathetic agent suggests that the empathetic agent was able to create a more positive and engaging interaction experience for participants. This is likely due to the fact that the empathetic agent was able to better understand and respond to the emotional needs of participants.

6.3. Neurophysiological responses

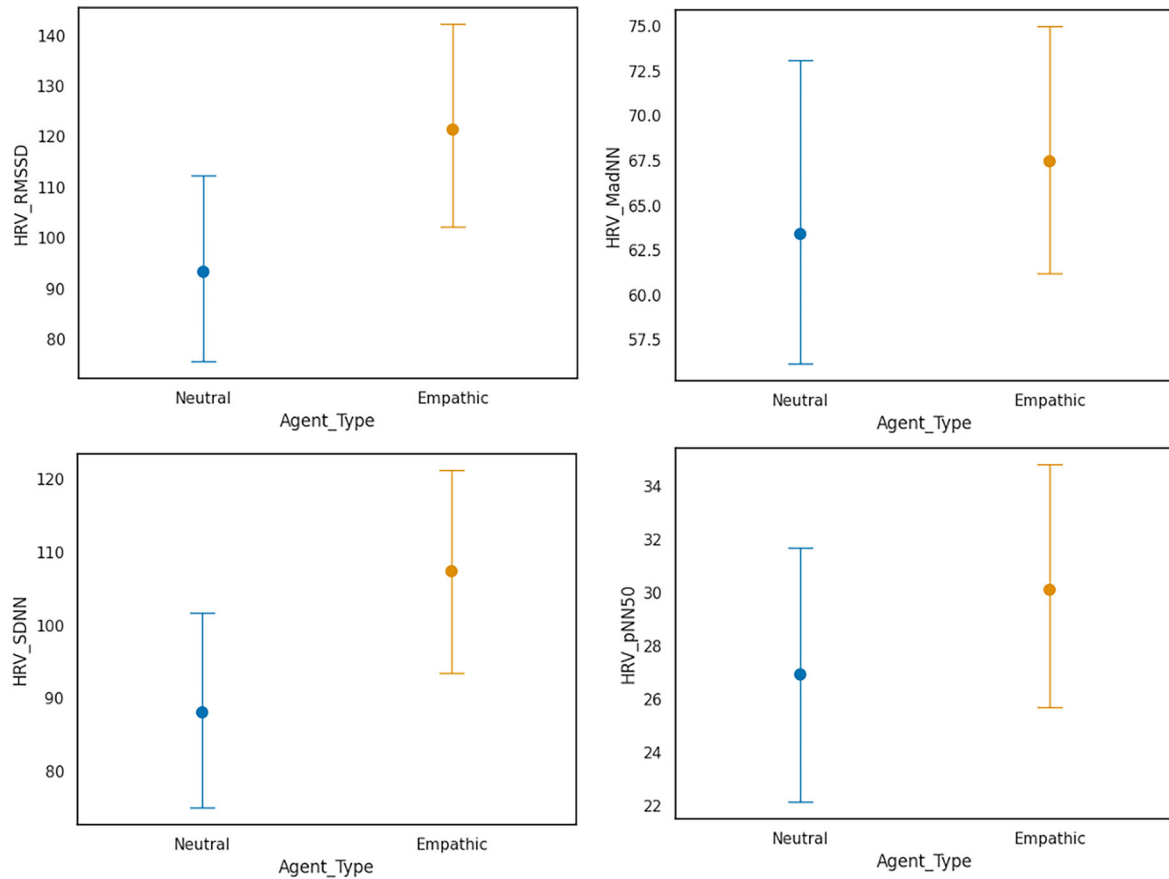
6.3.1. EEG signals

The ANOVA test results indicated that the agent type did not significantly impact EEG features. One possible reason for this lack of significance could be the presence of considerable noise in the EEG signals caused by talking during the conversation. The high levels of noise might have made it challenging for the statistical model to identify any discernible patterns in the data. Furthermore, during a conversation, multiple activities such as decision-making, language processing, motor cortex activity, emotion processing, and memory and information processing occur. These activities are likely to be common during interactions with both types of virtual agents. As a result, any subtle differences in brain activity related to the agent type may have been overshadowed by the shared neural processes.

Future research could focus on developing more robust noise reduction methods for EEG signals to better explore the effect of agent type on brain activities. In addition, using more detailed features in the analysis could help uncover subtle variations in brain activity that might be associated with different types of virtual agents. By addressing these challenges, we can gain deeper insights into the neural mechanisms underlying human-agent interactions and their potential implications for improving virtual agent design and usability.

Table 6. The result of statistical analysis for significantly different features for EDA and PPG signal in interaction with different agents.

Modality	Variable	<i>F</i>	Dof	Dof.res	<i>p</i>
EDA	peaks_var	9.12	1	111	<0.004
	peaks_std	9.12	1	111	<0.004
	peaks_mean	5.71	1	111	<0.020
	peaks_pos_derivatives	8.23	1	111	<0.005
	tonic_var	6.78	1	111	<0.020
PPG	tonic_std	6.78	1	111	<0.020
	HRV_CVNN	11.14	1	103	<0.002
	HRV_SDNN	10.82	1	103	<0.002
	HRV_RMSSD	9.98	1	103	<0.003
	HRV_SDSD	9.92	1	103	<0.003
	HRV_CVSD	9.29	1	103	<0.003
	HRV_HTI	8.89	1	103	<0.004
	HRV_TINN	8.44	1	103	<0.005
	HRV_MadNN	6.25	1	103	<0.020
	HRV_MedianNN	5.48	1	103	<0.030
	HRV_MCVNN	5.16	1	103	<0.030
	HRV_IQRNN	4.76	1	103	<0.040
	HRV_MeanNN	4.13	1	103	<0.050
	HRV_pNN50	4.06	1	103	<0.050

**Figure 7.** Point-plots of HRV features extracted from PPG signals.

6.3.2. PPG signals

The ANOVA test results revealed significant differences in almost all heart rate variability (HRV) features during interactions with neutral and empathetic DH (Table 6). The point plots in Figure 7 show participants exhibited higher levels of each HRV feature in interactions with the empathetic DH, indicating greater HRV. Figure 7 shows the point plot of the most important HRV features, and the trend of others is the same.

HRV is linked to enhanced emotional well-being (Mather & Thayer, 2018). It is also connected to reduced levels of stress, lower anxiety (Chalmers et al., 2014), and improved regulation of emotional

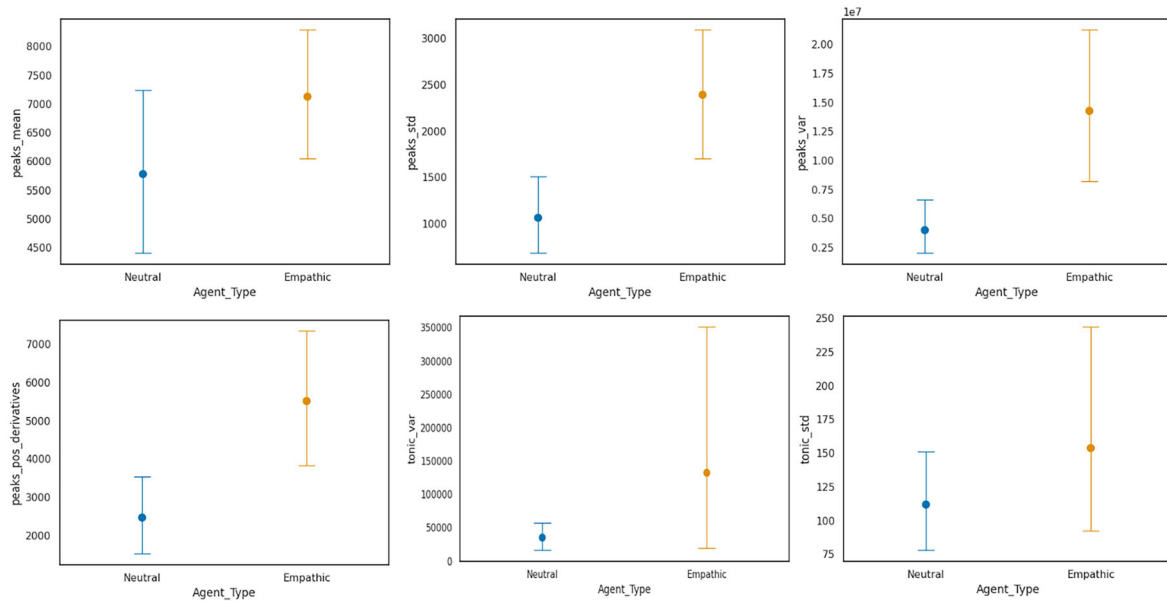


Figure 8. Point-plots of EDA features.

responses (Appelhans & Luecken, 2006) and empathy (Jauniaux et al., 2020). Therefore, the interactions with the empathetic agent appear to have helped individuals regulate their emotions more effectively than with the neutral DH. In addition, higher HRV is linked to increased attention (Appelhans & Luecken, 2006), suggesting that the empathetic agent was more engaging and effectively captured participants' attention.

6.3.3. EDA signals

The EDA analysis results revealed that the type of agent had a significant impact on the EDA signals, leading to differences in the statistical characteristics of EDA peaks and the tonic component (Table 6). As depicted in Figure 8, the average amplitude of EDA peaks was higher in interactions with the empathetic agent. A higher EDA peak amplitude indicates a greater increase in skin conductance, which is associated with higher emotional arousal (Benedek & Kaernbach, 2010). Additionally, the higher positive derivatives of EDA peaks observed in interactions with the empathetic agent suggest that the skin conductance is increasing at a faster rate during these interactions, further indicating higher emotional arousal.

While EDA signals may not directly reflect the valence state (Melander et al., 2017), they can serve as an indicator of emotional arousal, excitement, and engagement (Kreibig, 2010). In the current study, interactions with the empathetic agent resulted in an increase in skin conductance responses. This could be attributed to a more joyful and engaging experience during interactions with the empathetic agent, leading to heightened emotional arousal.

However, it is important to note that some participants provided post-study verbal feedback indicating feelings of nervousness while interacting with the empathetic agent. They found the concept of empathy emanating from a machine to be peculiar and unappealing. For example, some of them mentioned, "The empathetic condition was weird and I could not be able to connect with it. I usually expect a neutral reaction from a stranger or a machine," or "Empathetic Condition was realistic but weird." These feelings might have contributed to the observed increase in skin conductance responses during these interactions.

6.4. Real-time expressions

Throughout the interactions with the DH, three factors were collected to evaluate the emotions expressed by the DH in response to the human participants. These factors focused on assessing the appropriateness of reflected emotions, the expression of emotions, and the timing of these expressions.

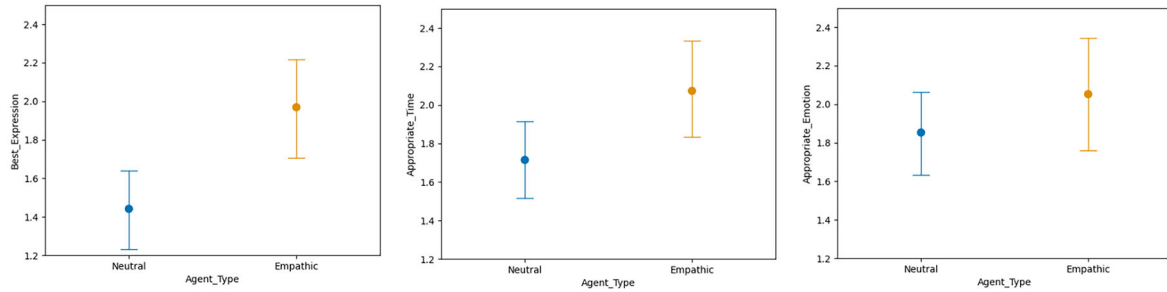


Figure 9. Effect of agent type on the appropriateness of emotion.

Table 7. Effect of emotional state on the appropriateness of emotion.

Variable	Condition	<i>F</i>	<i>Df</i>	<i>Df.res</i>	<i>p</i>
Appropriate emotion	Arousal	13	1	84	<0.001
Best expression	Arousal	6	1	83	<0.01
Best expression	Valence	6	1	86	<0.01
Appropriate time	Valence	9	1	85	<0.003
Appropriate emotion	Valence	12	1	86	<0.001

To understand the impact of the agent type on these factors, a set of one-way repeated-measure ANOVA with ART was performed. The result of statistical analysis demonstrated that the agent type significantly affected the appropriateness ($F(1, 166) = 10, p < 0.002$) and timing ($F(1, 166) = 6, p < 0.01$) of the reflected expressions.

In [Figure 9](#), the point plot illustrates the effect of agent type on the appropriateness of real-time emotions, expressions, and their timing. It is evident from the figure that the empathetic agent tended to use reflected expressions that were more appropriate and at the correct timing. Moreover, the reflected emotion quality was better in the empathetic DH, although this improvement did not reach statistical significance.

To explore the appropriateness of empathetic DH expressions in different emotional states, a series of two-way repeated-measure ANOVA with ART was conducted on participants' ratings in the empathetic condition, with arousal and valence as the factors. To consider the individual's actual emotional state during the conversation, we mapped the reported arousal and valence using [Equations \(1\) and \(2\)](#), respectively. These values served as the emotional state conditions in the statistical analysis.

The results revealed significant differences in the empathetic DH's effectiveness in displaying appropriate emotions, expressions, and timing across different valence levels. The empathetic DH's effectiveness in displaying appropriate emotions and expressions across arousal levels was also significant. However, there were no significant interactions between arousal and valence levels concerning emotion, expression, and timing appropriateness ([Table 7](#)).

The point plots presented in [Figure 10](#) show that the DH was more successful in expressing positive and high-arousal emotions. One possible explanation could be the model's proficiency in detecting high arousal and valence emotions. Another potential factor might be that the agent struggled to express negative emotions effectively.

6.5. Emotion recognition

We recorded the detected arousal and valence states using the EEG, PPG, and EDA signals and their fusion in real-time. Then, we compared them with the reported arousal and valence levels. Since the reported labels were based on a 5-point Likert scale, we converted them to low(−1) and high(1) according to [Equations \(1\) and \(2\)](#).

[Table 8](#) shows the percentage of how well the recognized arousal and valence levels using various modalities match the reported binary arousal and valence levels. It shows the percentage of correct recognition in real-time using various modalities compared to the reported binary arousal and valence

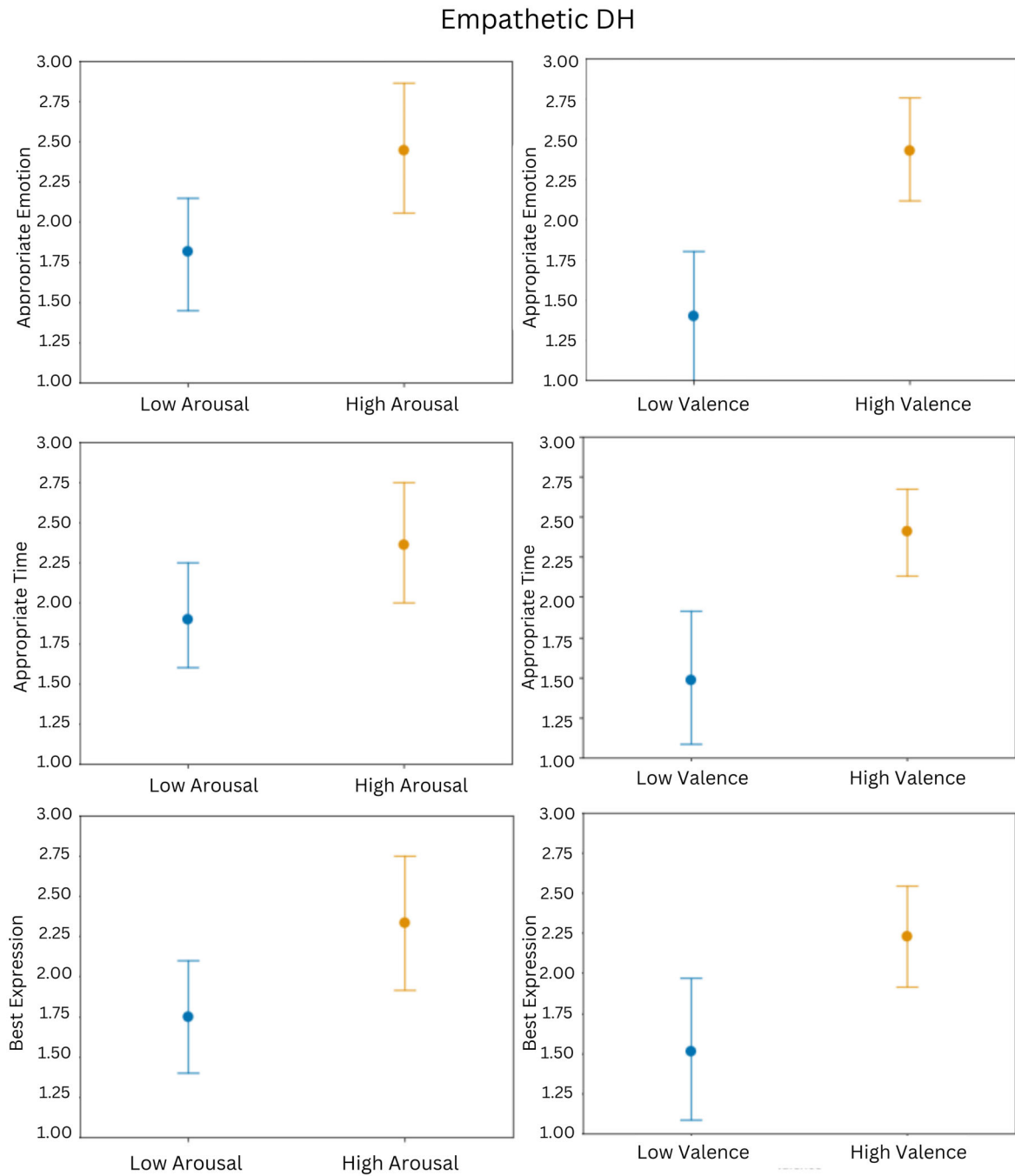


Figure 10. Point-plots of the effect of emotional state on the appropriateness of emotion, expression, and timing in the empathetic DH.

Table 8. The percentage of alignment between the recognition of various modalities and the reported arousal and valence levels.

	Arousal	Valence
EEG	64.8	57.1
PPG	65.4	58.1
EDA	58.2	52.7
Fusion	69.1	57.3

Note: The bold values highlights the accuracy of fusion method which has been used for the recognition.

levels. It is evident that the percentage of similarity between the reported emotion and the emotion used for generating real-time expressions and selecting appropriate responses using the fusion method is higher compared to other modalities for both arousal and valence levels. Despite these significant

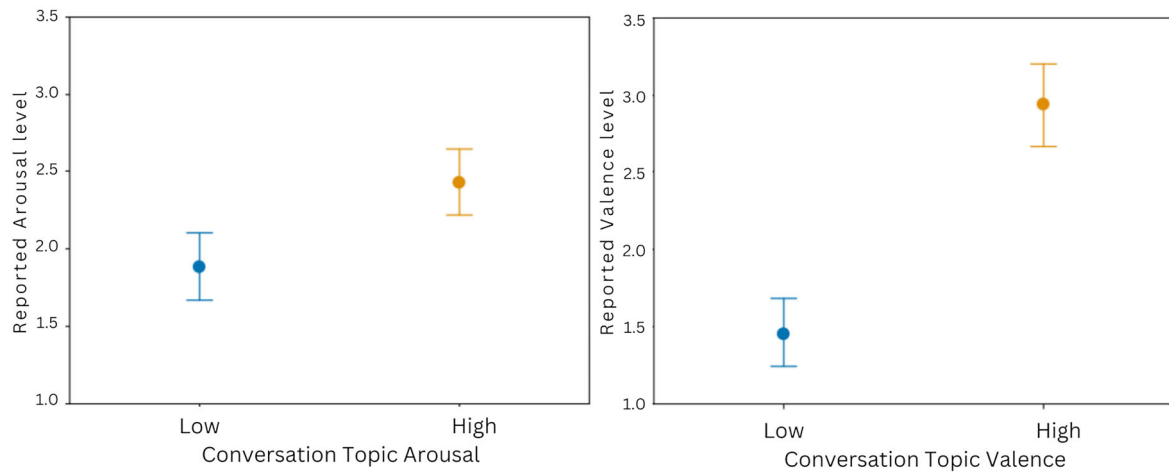


Figure 11. Comparing the reported arousal and valence levels with the emotional state of the conversation topics in order to estimate the effectiveness of inducing emotion. In these two point-plots, the x-axis shows the emotional state (arousal or valence level) of the conversation topic, and the y-axis shows the average of the reported emotional state based on a 5-point Likert scale. It shows a good level of alignment between the conversation topic and emotional state.

percentages, future improvements in recognition results can be achieved by employing more robust models.

6.6. Emotional state

We conducted an assessment by comparing the reported arousal and valence levels with the emotional state of the conversation topics in order to estimate the effectiveness of conversations in inducing emotions. To achieve this, we utilized a one-way repeated-measure ANOVA with ART, which allowed us to examine the influence of the emotional state of the conversation on individuals' emotions. The results indicated that the emotional state of the conversation was significantly affected by the target emotion of the conversation (p -value < 0.05). As shown in Figure 11, the depicted data reveals that the reported arousal and valence ratings correspond closely with the emotional state of the conversation topics. This indicates that the conversations were effective in eliciting emotions.

7. Discussion

In this paper, we created a real-time emotion recognition application that recognizes emotions using multimodal data. We combined neural and physiological signals to recognize emotion every 5 s in real-time settings and integrated it with a DH from the Soul Machines company. The real-time emotions fed into the DH, and DH created its expressions and responses according to the detected emotions empathetically by mirroring the participant's emotional state according to Table 1. Using the multimodal real-time emotion recognition system and the integrated DH, we designed a user study and compared human neurophysiological responses and self-reports in empathetic and neutral mode.

To evaluate the effectiveness of the neurophysiological emotion recognition and mirroring system in enhancing human-agent empathy, we assessed cognitive, affective, and somatic components of empathy. Self-reported measures revealed significantly higher levels of overall empathy, cognitive empathy (emotion recognition), and affective empathy (emotional expression) during interactions with the empathetic digital human (DH) compared to the neutral version. Although average empathy scores were moderate, they were strongly modulated by the participant's emotional state, with higher empathy reported during positive and high-arousal experiences.

Physiological measures, including heart rate variability (HRV) and electrodermal activity (EDA), provided additional support for these findings. Interactions with the empathetic DH resulted in significantly higher heart rate variability (HRV) and electrodermal activity (EDA), indicating increased emotional arousal, engagement, and improved emotional regulation. EEG signals, however, did not show significant differences, likely due to conversational noise and overlapping cognitive processes.

Real-time emotion expressions by the DH were rated as more appropriate and better timed in the empathetic condition, particularly during high-arousal and positive-valence states. Although emotion recognition accuracy was modest (69% for arousal, 57% for valence), the fusion method outperformed individual modalities and enabled reasonable real-time responses. Additionally, the conversational topics effectively elicited target emotional states, validating the emotion induction protocol.

Our statistical analysis of PPG and EDA signals also revealed a significantly higher level of HRV and skin conductance responses in interaction with the empathetic agent. These findings show that people experienced more emotional arousal and engagement in interaction with the empathetic agent. They also had better emotion regulation in interaction with the empathetic agent than with the neutral agent. Although our analysis of EEG signals showed no significant effect of agent type on EEG features. This may be due to substantial noise introduced by speaking during the interaction, which likely masked any subtle differences in neural activity. Additionally, shared cognitive processes—such as language comprehension, decision-making, motor activity, and emotion processing—are common across interactions with both agents and may have overshadowed agent-specific effects. Future studies could explore more advanced noise reduction techniques and include richer EEG features to better capture nuanced neural differences associated with agent type.

As this study was exploratory, multiple univariate analyses were conducted to identify broad patterns rather than confirm specific hypotheses. While we acknowledge the concern regarding multiple comparisons and false-positive rates, applying strict corrections (e.g., Bonferroni (Dunn, 1961)) could suppress meaningful insights. Future research should consider multivariate approaches (e.g., PCA (Maćkiewicz & Ratajczak, 1993)) or hypothesis-driven statistical adjustments to enhance robustness while maintaining sensitivity to emerging trends.

Overall, participants reported greater enjoyment, engagement, and empathy during interactions with the empathetic digital human (DH). These findings highlight the potential of neurophysiological emotion recognition in shaping more empathetic agents. However, while promising, the results also point to areas that need further refinement. The following section highlights the key challenges and limitations of our research and future directions.

- **Sample Size:** One key limitation of this study is the relatively small sample size. However, our sample size was informed by previous research in EEG-based emotion recognition and affective computing, where small to moderate samples are common, particularly in exploratory, technically demanding studies (Koelstra et al., 2012; Saffaryazdi et al., 2024). Given the technical complexity of our setup, including real-time emotion recognition based on physiological and EEG data during live interaction with an expressive virtual agent, our recruitment capacity was inherently limited. Furthermore, our primary focus in this initial study was on developing and validating the real-time emotion mirroring pipeline rather than conducting large-scale statistical inference. Future research should address this limitation by performing a priori power or sensitivity analyses and recruiting larger, more diverse samples to support more robust and generalizable conclusions.
- **Recognition Accuracy:** The level of empathy between humans and the digital human agent was directly influenced by the accuracy of the emotion recognition method. However, achieving a robust model capable of accurately and swiftly detecting emotions using neurophysiological signals requires further exploration and research. In the future, we can enhance recognition accuracy by using deep learning methods and adopting a more robust fusion strategy. It is essential to also consider the recognition speed while seeking to improve accuracy. Incorporating behavioral modalities such as facial expressions and speech recognition could also offer the benefits of both behavioral and neurophysiological modalities. By integrating these modalities, we can potentially enhance the overall performance and effectiveness of the empathetic conversational agent.

- **Avatar Diversity:** We used a pre-configured female avatar that was culturally familiar and technically compatible, which helped ensure consistency across participants. However, relying on a single avatar in terms of gender, age, and cultural appearance may limit the generalizability of the findings. People may respond differently depending on the avatar's characteristics. For example, some may feel more at ease with females, while others may relate better to males (Brophy, 1985; Tam et al., 2020). Due to recruitment and technical constraints, we focused on one avatar in this initial study. Future work should explore a broader range of avatars to better understand how demographic factors influence user experience and empathy.
- **Emotional Transition:** Different modalities reflect the emotional transition on different time scales. For instance, the EEG signal can swiftly predict emotional transitions, while the PPG signal takes some time to accurately reflect the real emotion, resulting in differing recognition outcomes across modalities. Although this study did not extensively explore emotional transitions, such transitions commonly occur in everyday conversations. Further research is required to investigate the impact of emotion transitions on different modalities and their recognition accuracy to better understand and leverage their potential in empathetic conversational agents.
- **Personality and Trait Empathy Effects:** Individuals with different personality traits and baseline levels of empathy may experience varying degrees of connection with the same virtual agent, even in similar situations. Some participants might feel awkward or uncomfortable interacting with a machine, while others may feel more relaxed and open compared to interacting with a human. These individual differences can influence how users perceive and engage with emotionally responsive agents. To improve the reliability and generalizability of the findings, future studies should aim to include a larger and more diverse participant pool and assess trait empathy using a standardized tool such as the Toronto Empathy Questionnaire (TEQ) (Spreng et al., 2009) to reduce the potential influence of personality variations on the results.
- **Voice Tone:** A key limitation of this study is the lack of vocal prosody control in the digital human (DH) used. Although tone of voice is a fundamental component of empathic communication, the version of the Soul Machines' DH implemented here could not modulate its vocal tone to reflect its emotional state. This omission may have weakened the overall empathic impact of the interaction, as participants may have perceived a mismatch between verbal content and emotional expression. Future iterations of the system should incorporate agents capable of dynamically modulating vocal prosody in real time, aligned with the user's detected emotional state. This addition could enhance the perceived authenticity and warmth of the DH's responses and contribute to a richer, more engaging empathic experience.
- **Language model:** The version of Soul Machines' Digital Human (DH) used in this study did not support integration with Large Language Models (LLMs). Consequently, the agent's verbal output was limited to a predefined set of responses matched to detected emotional states, without real-time conversational flexibility. While this limitation did not affect our core focus—emotional mirroring through nonverbal facial expressions—it constrained the richness of verbal interaction. As part of our research, we are exploring newer versions of the platform that incorporate LLM capabilities. Future integration of LLMs could enable more dynamic, context-aware dialogue, thereby enhancing user engagement and further supporting empathic human-agent interaction.
- **Calibration Trade-offs in Neurophysiological Data:** This study did not include a calibration step for individual participants, despite the use of external equipment such as the OpenBCI cap, which captures neurophysiological signals known to vary across individuals. Instead, we relied on a general emotion recognition model and applied data normalization techniques to reduce inter-participant variability and ensure signal comparability. General models offer the advantage of scalability and reduced setup time, making them suitable for real-world applications where individualized calibration may be impractical. While this approach was sufficient for identifying population-level patterns and evaluating the overall system functionality, it may have limited the model's ability to capture individual-specific emotional responses. Future research could investigate whether incorporating a personalized calibration phase enhances emotion recognition accuracy and improves the perceived empathy of the virtual agent.

8. Conclusions

In this study, we developed a real-time emotion recognition system that continuously detects users' emotional states every five seconds using multimodal neurophysiological data—specifically EEG, EDA, and PPG signals. These detected emotional states were used to drive the behavior of a digital human (DH) developed by Soul Machines. The DH mirrored participants' emotional states through facial expressions and verbal responses, as defined in Table 1, enabling more empathetic interactions. By integrating this system, we conducted a user study to compare human neurophysiological responses and self-reported experiences during interactions with both empathetic and neutral versions of the DH. This work served as a first step in validating the feasibility of a neurophysiology-driven empathic agent, aiming to enhance human-machine empathy without relying on traditional audio-visual emotion inputs.

The findings indicate that participants experienced stronger emotions and engagement during interactions with empathetic agents, as evidenced by neurophysiological responses and self-reports. However, challenges and limitations were also encountered, highlighting areas for improvement in future research.

Key areas for enhancement include using large language models (LLMs) for making the conversations and improving the accuracy of emotion recognition methods. LLMs, such as GPT-based models, can generate more natural, context-aware, and engaging dialogue. By incorporating these models, we can simulate human-like conversations that adapt dynamically to user input, making interactions feel more authentic and engaging.

Increasing the sample size and using deep learning techniques can potentially improve the accuracy of emotion detection. In addition, integrating behavioral modalities such as facial expression analysis and speech recognition enriches the understanding of user emotions. For example, combining data from facial expressions, vocal tone, and textual content can provide a more comprehensive emotional profile. This holistic approach ensures the system can detect and respond to emotions even when one modality is less informative or unavailable. However, we should consider the limitations of each modality in all situations.

Incorporating voice tone modulation capabilities into digital human agents could considerably enhance the empathetic interaction experience, addressing a current limitation observed in the study.

Overall, while the findings demonstrate the potential of real-time emotion recognition and empathetic digital agents, ongoing research is essential to address the identified challenges and further refine the development of empathetic conversational agents. By addressing these limitations, future studies can

Notes

1. <https://pegconv.nastaran-saffar.me#pegmconv>.
2. <https://www.soulmachines.com/>.
3. <https://www.metahuman.com>.
4. <https://openbci.com>.
5. <https://shimmersensing.com>.

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Author contributions

CRedit: Nastaran Saffaryazdi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Tamil Selvan Gunasekaran:** Investigation; **Kate Loveys:** Conceptualization, Methodology; **Elizabeth Broadbent:** Conceptualization, Methodology, Supervision, Writing – review & editing; **Billinghurst Mark:** Conceptualization, Methodology, Resources, Supervision, Validation, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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